

Total No. of Questions : 10]

SEAT No. :

P2993

[5154]-558

[Total No. of Pages : 3

B.E. (Mechanical)

FINITE ELEMENT ANALYSIS

(2012 Course) (402050B) (Elective - IV) (Semester - II)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Draw suitable neat diagrams, wherever necessary.*
- 2) *Figures to the right indicate full marks.*
- 3) *Use of electronic pocket calculator is allowed.*
- 4) *Assume suitable data, if required.*

Q1) a) Explain importance of Boundary conditions and further explain: **[6]**

- i) Essential Boundary Conditions.
- ii) Natural Boundary Conditions.

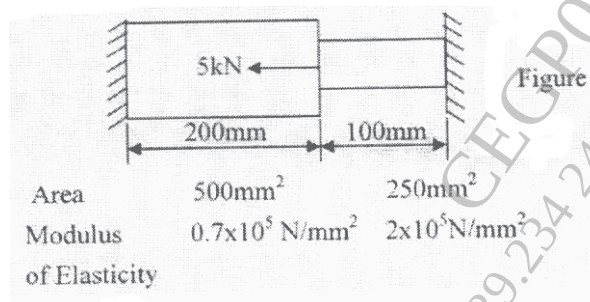
b) Why quality of meshing is important in FEA and how it is ensured and how the convergence of element in FEA formulation is achieved? **[4]**

OR

Q2) a) Explain the principle of Galerkin's Weighted Residual Method. **[6]**

b) Write a step by step procedure for Weak Formulation in elemental formulations. **[4]**

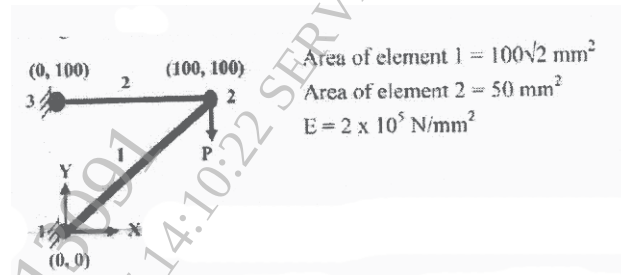
Q3) Determine the nodal displacements and element stresses by finite element formulation for the following figure. **[10]**



OR

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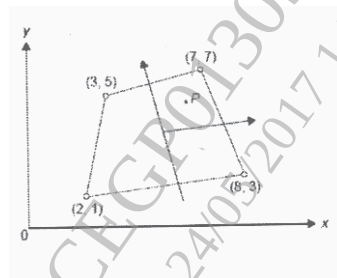
- Q4) a)** For the truss shown in figure acted by a force P. the displacements of node 2 is found to be $U_x = 0.01 \text{ mm}$ and $U_y = -0.02 \text{ mm}$, Determine the Force P. [6]



- b) Explain method of Penalty for solution of equation $[K] \{X\} = \{F\}$ where, $[K]$ is a stiffness matrix, $\{X\}$ is displacement vector and $\{F\}$ is load vector. [4]

- Q5) a)** What is meant by ISO, SUPER and SUB parametric element and for structural analysis which is mostly preferred? [6]

- b) Determine the Cartesian coordinate of the point P ($\zeta=0.5, \eta=0.5$) shown in Fig. [6]



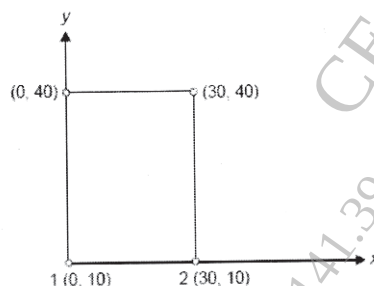
- c) State and explain the three basic laws on which isoparametric concept is developed. [6]

OR

- Q6) a)** Write short notes on: [8]

- Uniqueness of mapping of isoparametric elements.
- Jacobian matrix.

- b) For the element shown in Fig. assemble Jacobian matrix and strain displacement matrix for the Gaussian point (0.7, 0.5). [10]

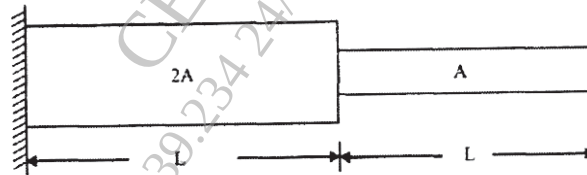


- Q7) a)** Write down governing equation of steady state Heat Transfer and also write down elemental stiffness matrix and compare with Bar element. [6]
- b)** Consider a brick wall of thickness 0.6m, $k = 0.75 \text{ W/m}^\circ\text{K}$. The inner surface is at 15°C and the outer surface is exposed to cold air at -15°C . The heat transfer coefficient associated with the outside surface is $40 \text{ W/m}^2^\circ\text{K}$. Determine the steady state temperature distribution within the wall and also the heat flux through the wall. Use two elements and obtain the solution. [10]

OR

- Q8) a)** Heat is generated in a large plate ($K = 0.5 \text{ W/m}^\circ\text{C}$) at the rate of 2000 W/m^3 . The plate is 10 cm thick. Outside surface of the plate is exposed to ambient air at 30°C with a convective heat transfer coefficient of $40 \text{ W/m}^2^\circ\text{C}$. Determine the temperature distribution in the wall. [10]
- b)** Derive FEA stiffness matrix for Pin Fin Heat Transfer Problem. [6]

- Q9) a)** Write down Consistent Mass and Lumped Mass Matrix for [6]
- Bar Element
 - Plane Stress Element
- b)** Find the natural frequencies of longitudinal vibrations of the same stepped shaft of areas $A = 12000 \text{ mm}^2$ and $2A = 2500 \text{ mm}^2$ and of equal lengths ($L = 1\text{m}$), when it is constrained at one end, as shown below: [10]



OR

- Q10) a)** Explain difference between consistent and lumped mass matrix technique for modal analysis of structure. [6]
- b)** Find the natural frequencies of longitudinal vibrations of the unconstrained stepped shaft of areas A and $2A$ and of equal lengths (L), as shown below: [10]

